



Role of Plant Health Parameters in Understanding Spatial Heterogeneity of Hydraulic Conductivity of Vegetated Soil: A Case Study of Urban Green Infrastructure Monitoring

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Abstract. Development of green infrastructure is widely adopted as a key strategy for enhancing socio-ecological benefits in urban areas. Generally, four categories of vegetation exist in urban green infrastructure. Those are (i) vegetated soil under tree shade during entire daylight period, (ii) vegetated soil under tree shade during 3–4 h of daylight period, (iii) vegetated soil under light during entire daylight period and (iv) soil surface covered by mix grass and shredded leaves. Previous studies have shown that presence of vegetation may influence hydraulic conductivity. The main factors those govern such influence are found to be growth of vegetation, which is directly related to photosynthesis and stomatal conductance. Stomatal conductance is the measure of passage of carbon dioxide or water vapor through stomata of leaf. The objective of this study is to investigate role of plant health parameters in understanding spatial heterogeneity of hydraulic conductivity in urban green infrastructure. Field monitoring of mix vegetated soil was conducted for about three months. Plant health is investigated in terms of vegetation growth and stomatal conductance. Stomatal conductance and hydraulic conductivity were measured in 150 locations in selected site once every month. Stomatal conductance was measured using an electronic sensor (leaf porometer). Mini disk infiltrometer was used to measure hydraulic conductivity. Stomatal conductance of mix grass under light for longer duration (around 8 h) was found to be higher than that under light for shorter duration (3–4 h) and shade. Hydraulic conductivity of mix grass cover under shade was found to be relatively low. As compared to stomatal conductance, preferential flow is found to be more dominant in governing the hydraulic conductivity.

Keywords: Stomatal conductance · Vegetation growth
Hydraulic conductivity

1 Introduction

Hydraulic conductivity of vegetated soil is important property that governs runoff, ground water recharge and slope stability [1, 2]. It is also vital to understand the long-term performance of sustainable drainage systems (i.e., urban green infrastructure; [3]). Suction induced in root zone [4] due to evaporation or evapotranspiration and preferential flow around the root govern the hydraulic conductivity. It is evident that hydraulic conductivity of vegetated soil is affected by vegetation density and shredded leaves [5]. Previous studies show that hydraulic conductivity of vegetated soil could be higher [6] or lower [7] than that of bare soil. In addition, it was reported by Leung et al. [2] that, hydraulic conductivity may vary for different vegetation depending on the stomatal conductance. It can be explained by using Monteith modification of Penman equation [2]. Higher stomatal conductance induces greater evapotranspiration. Greater evapotranspiration induces higher suction [2]. Such higher suction implies less hydraulic conductivity [1].

Mix vegetated soil is found to widely exist in the urban areas as part of green infrastructure (lawns, biofilters and green roofs) to provide socio-ecological benefits [8] and also storm water management [9]. Generally four categories of vegetated soil occur in mix vegetated soil. Those are (i) vegetated soil under tree shade during entire daylight period, (ii) vegetated soil under tree shade during 3–4 h of daylight period, (iii) vegetated soil under light during entire daylight period and (iv) soil surface covered by mix grass and shredded leaves. Gadi et al. 2017 shows that, hydraulic conductivity through these four categories could be different. However, stomatal conductance was not measured in their study. In addition, the hydraulic conductivity tests were conducted only once after vegetation density is reached to 100%. Further growth of vegetation may increase or decrease the hydraulic conductivity due to changes in (1) preferential flow and (2) variation in soil suction. Vegetation density (m^2/m^2) is defined as projected area of vegetation to per unit area [5]. Furthermore, influence of stomatal conductance and vegetation growth was rarely investigated collectively. Such collective investigation is vital to understand the spatial variation of hydraulic conductivity in urban green infrastructure and storm management. The objective of this study is to investigate the influence of plant health parameters on spatial heterogeneity of hydraulic conductivity in mix vegetated soil (i.e., urban green infrastructure).

2 Materials and Methods

2.1 Site Description

Field testing site was selected within campus of Indian Institute of Technology (IIT) Guwahati, which is located in North Eastern part of India near banks of River Brahmaputra. Mixed vegetation consists of *Pongamia Pinnata* tree and several grass species. Mix grass cover contains *Poacea* and *Bauhinia purpurea* species. Field monitoring was conducted on mix grass cover to better understand the spatial variation of stomatal conductance, and hydraulic conductivity.

2.2 Soil Properties

Four disturbed soil samples were collected from different locations at testing site. In situ dry density at four locations was observed to vary between 1315 kg/m^3 and 1387 kg/m^3 , with an average value of 1351 kg/m^3 (approximately equal to 78% of dry density). The gravel, sand, and silt and clay contents were found to be 0%, 98.6% and 1.4%. Based on the measured particle size distribution, the soil in the selected site is classified as poorly graded sand (SP; [10]), according to unified soil classification system (USCS). Saturated hydraulic conductivity was found to be $3.7 \times 10^{-4} \text{ m/s}$.

2.3 Overview of Testing Site Consisting Mixed Vegetation

Poacea and *Bauhinia purpurea* species were selected based on (i) broad availability in sub-tropical regions and (ii) drought tolerance ability, which is preferable for bio engineered slope stabilization [5]. *Pongamia pinnata* is selected based on its wide spread presence in sub-tropical regions. It was recognized as the landscape resource [5]. Area of mix grass cover selected for the present study is marked in Fig. 1a ($6 \text{ m} \times 5 \text{ m}$). Non-uniformity in shoot length of vegetation in the study site can be seen in Fig. 1a.

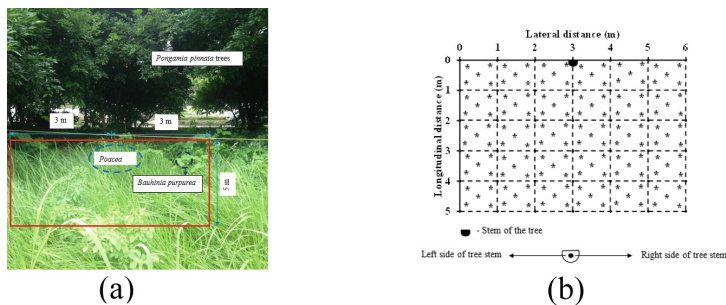


Fig. 1. (a) Overview of mix vegetation in test site and (b) Categorization of selected site into small grids for quantifying spatial variation of stomatal conductance and hydraulic conductivity

2.4 Instrumentation on the Vegetated Soil

Typical layout showing locations (150 measurements), where stomatal conductance and hydraulic conductivity were measured is shown in Fig. 1b. The selected area in the field is categorized into small grids for measuring stomatal conductance and hydraulic conductivity. The selected grid size was determined based on trial measurements of stomatal conductance and hydraulic conductivity. Stomatal conductance was measured using leaf porometer. Leaf porometer [11] allows the leaf in contact with the two known conductance elements. The stomatal conductance is estimated from the difference in humidity between leaf, two sensors and desiccant [12]. Hydraulic conductivity in mix grass cover was measured using mini disk infiltrometer (MDI; [13]). Ghestem et al. [14] showed that, diameter of fine root may decrease and become 40% of its

original diameter, when radiant energy is maximum. Hence, significant preferential flow occurs through soil-root interface during afternoon. Hydraulic conductivity measurements were performed during afternoon to take the preferential flow into account.

2.5 Field Monitoring Programme

The field monitoring programme for quantifying the stomatal conductance and vegetation growth influence on hydraulic conductivity in mixed vegetation site was conducted from 1st May 2017 to 31st July 2017. It is evident that, stomatal conductance and hydraulic conductivity are quite uncertain spatially. To quantify the spatial uncertainty, stomatal conductance and hydraulic conductivity were measured at 150 pints (refer to asterisk (*) in Fig. 1b) once every month. Previous studies show that, stomatal conductance can be influenced by leaf age [5]. Hence, leaves of approximately similar age were considered for the present study. The mix grass growth in the site was started in April month. The mix grass exist on the site in the previous months was wilted due to season change. In addition, tree leaves were also shredded due to season change. The grass shoots selected to measure stomatal conductance were marked for identification in field monitoring period. A series of hydraulic conductivity experiments were performed at designated locations. Hydraulic conductivity was measured using Zhang [15] approach. The Van Genuchten parameters (n and α) were adopted from Carsel and Perish [16] for calculation, which are established using soil water retention curve of bare soil. However, these parameters may not be same for bare soil and vegetated soil [2, 5]. As the main focus of present study is to improve the working knowledge on hydraulic conductivity spatial variation, Van Genuchten parameters of bare soil were adopted. Rainfall depths during May, June and July were found to be 214 mm, 326 mm and 285 mm, respectively. This shows that the tests were conducted during wet period.

3 Results and Discussions

3.1 Vegetation Cover Change During Monitoring Period

Figure 2a, b and c shows the overview of variation in surface area of the selected site during field monitoring period. Figure 2a shows the image captured on 31 May 2017. Typically, four categories of mix grass can be observed during monitoring period (Marked in Fig. 2a). Those are (a) mix grass under tree shade during entire daylight period, (b) mix grass under tree shade during 3–4 h of daylight period, (c) mix grass under light during entire daylight period and (d) surface covered by mix grass and shredded leaves. The proportions of these four categories of mix grass were found to vary with time. Majority area of site is found to cover with *Poacea* during May. *Bauhinia purpurea* is also observed in relatively small area. Figure 2(b) and (c) shows the images of mix grass cover at the end of June 2017 and July 2017. *Poacea* and *Bauhinia purpurea* are found to keep on growing during June and July. Significant area covered by *Bauhinia purpurea* was observed at the end of July. In addition, *Pongamia*

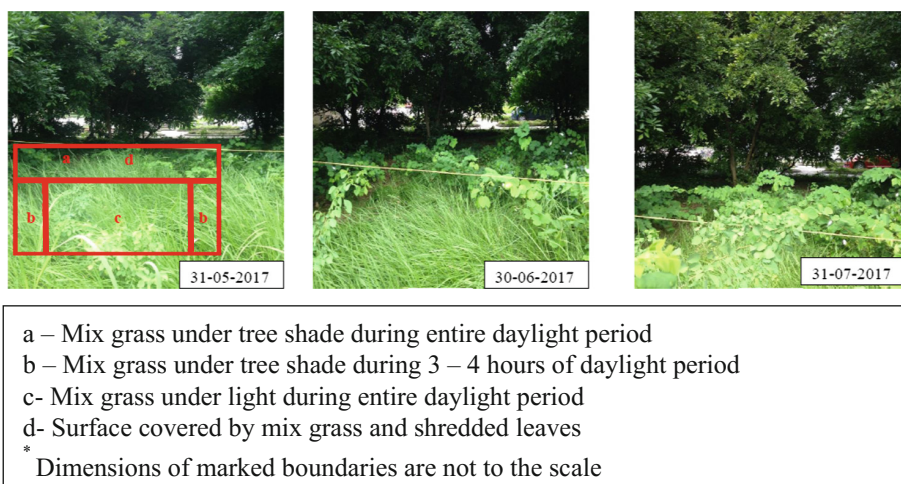


Fig. 2. Mix vegetated soil at the end of three different months: (a) 31 May 2017 (b) 30 June 2017 (c) 31 July 2017

pinnata plants were found to grow within 1 m longitudinal distance in left side of tree stem during June and July.

3.2 Spatial Variation of Stomatal Conductance

Figure 3a, c and e illustrates the spatial variation of stomatal conductance range for three months. Contours were used for illustrating the stomatal conductance. Stomatal conductance is found to vary between $42 \text{ mmol/m}^2\text{s}$ to $510 \text{ mmol/m}^2\text{s}$. Stomatal conductance was observed to be uncertain spatially. Figure 3a shows spatial variation of stomatal conductance at the end of May. Stomatal conductance of mix grass under light for longer duration (8 h) is observed to be higher than that under light for shorter duration (3–4 h) and shade. Furthermore, stomatal conductance is higher at greater longitudinal distance from tree stem as compared to that near the tree stem. This may be due to absence of shade at Spatial variation of stomatal conductance at the end of June is shown in Fig. 3c. An increase in stomatal conductance over entire area in the selected site was found during June. This is due to growth of *Bauhinia purpurea*, *Poacea* and *Pongamia pinnata* during June. Such growth indicates presence of young shoots, which implies relatively higher stomatal conductance [17]. Trend of variation of stomatal conductance with respect to four categories of mix grass at the end of June is similar to that at the end of May. Figure 3e shows spatial variation of stomatal conductance at the end of July. Unlike to June, stomatal conductance is observed decrease during July. One reason for this may be due to significant increase in *Bauhinia purpurea* during July. Such increase in *Bauhinia purpurea* implies raise in area of mix grass cover under shade. A difference of 17%–350% in stomatal conductance is found between mix grass under shade and light. Results reported by Gross et al. [18] also showed such difference could be up to 450% could be possible. This revealed to be attributed to the relatively low radiant energy under shade [18].

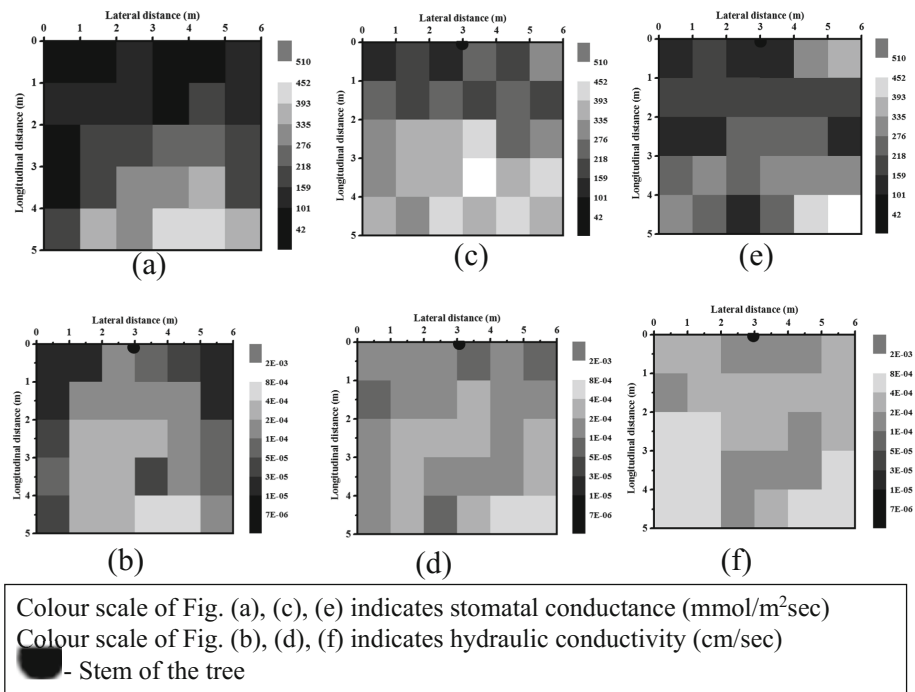


Fig. 3. Spatial variation of stomatal conductance and hydraulic conductivity ranges at the end of: (a) & (b) May 2017; (c) & (d) 30 June 2017; (e) & (f) 31 July 2017

3.3 Spatial Variation of Hydraulic Conductivity

Figure 3b, d and f illustrates spatial variation of hydraulic conductivity range in mix vegetated soil for three months. Range of hydraulic conductivity is found between 7×10^{-6} cm/s and 2×10^{-3} cm/s. Figure 3b shows spatial variation of hydraulic conductivity at the end of May. Hydraulic conductivity of mix grass cover under shade is found to be relatively low among the four categories of mix grass cover. Such low hydraulic conductivity may be attributed to presence of tree roots. Root water uptake through tree roots may induce relatively higher suction as compared to that by mix grass [4]. Higher suction implies less hydraulic conductivity. Although the vegetation densities in both side of tree stem are similar, hydraulic conductivity in right side of tree stem is higher than that in left side within 1 m longitudinal distance. Reason for this may be the presence of greater area covered by mix grass cover in right side of tree stem as compared to that in left side of tree stem, which implies higher preferential flow [5]. This reason may also be attributed to the difference between hydraulic conductivity of grass under light for 3–4 h in left side and right side of tree stem. Hydraulic conductivity through mix grass cover under light during entire daylight period is found to be highest among four categories of mix grass cover. One reason for this may be, higher preferential flow due to presence of mix grass in relatively greater area [5].

Figure 3d shows spatial variation of hydraulic conductivity at the end of June. An increase in hydraulic conductivity can be found in three categories mix grass cover (i.e., mix grass under tree shade, mix grass under tree shade and surface covered by mix grass and shredded leaves). Any increase in hydraulic conductivity through mix grass under light is not found. This may be due to enhanced root water uptake (and stomatal Fig. 3f shows the spatial variation of hydraulic conductivity at the end of July. An increase in hydraulic conductivity is found in three categories of mix grass cover (i.e., mix grass under tree shade during entire daylight period, mix grass under tree shade during 3–4 h of daylight period and surface covered by mix grass and shredded leaves). Any increase in hydraulic conductivity through mix grass under light during entire daylight period is not found. Unlike to previous months, hydraulic conductivity through mix grass within 2 m lateral distance is found to increase during July. Any increase in hydraulic conductivity through mix grass under light during entire daylight period is not found. It can be observed that, larger stomatal conductance exhibits relatively higher hydraulic conductivity through mix grass under light during daylight period. This shows that, preferential flow effect around roots dominates the effect of higher suction induced due to greater stomatal conductance. In addition, lower stomatal conductance exhibits relatively less hydraulic conductivity through mix grass under shade during entire day. This shows that, suction induced due to tree roots dominates the effect of stomatal conductance.

4 Conclusions

The case study explored the spatial variation of hydraulic conductivity in a natural urban green infrastructure containing soil vegetated with mixed species. Spatial variation of hydraulic conductivity was interpreted using plant health parameters (stomatal conductance and vegetation growth). Stomatal conductance was found to vary between $42 \text{ mmol/m}^2\text{s}$ to $510 \text{ mmol/m}^2\text{s}$. Whereas, hydraulic conductivity was found to vary between $7 \times 10^{-6} \text{ cm/s}$ and $2 \times 10^{-3} \text{ cm/s}$.

Stomatal conductance of mix grass under light for longer duration (during entire daylight period) is observed to be higher than that under light for shorter duration (3–4 h) and shade. Presence of young shoots (newly grown mix grass) increase the stomatal conductance. However, excess growth of vegetation accommodates shade on the mix grass, which reduces stomatal conductance. Hydraulic conductivity of mix grass cover under shade was found to be relatively low. This may be attributed to presence of tree roots. Hydraulic conductivity of mix grass cover increases under shade with growth. This may be due to raise in preferential flow with mix grass growth. Whereas, any increase in hydraulic conductivity with growth for mix grass under light during entire day was not found. As compared to stomatal conductance, preferential flow is found to be more dominant in governing the hydraulic conductivity. Further, interaction between plant health parameters and hydraulic conductivity among each category of mix grass needs to be explored. In addition, long term monitoring with consideration of various vegetation species and seasonal variation can be helpful.

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